LESSON 4.4

Find and Compare with Percentages

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the percent of a number using an equation or a model.

WHY THIS MATTERS It's Discount Day at the arcade! Different games are offering percent-off deals on ticket prices. The claw machine is 25% off its usual 60-ticket price, and the racing game is 40% off 80 tickets. You need to calculate the discounts to figure out how many tickets each game actually costs!

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Percent *Spanish: Porcentaje*	A way to compare a number to 100, shown with the % sign.	45% = 0.45
Base *Spanish: Base*	The whole amount you start with — the number right after the word "of."	45% of 300 → base is 300
Part *Spanish: Parte*	How much of the base (the whole) you get. It is your answer.	45% of 300 is 135
Equation *Spanish: Ecuación*	A number sentence with an equal sign. The percent one is: part = percent × base.	part = 0.45 × 300
Discount *Spanish: Descuento*	Money taken off the first price to make it cheaper.	$40 shirt at 25% off: discount = $10
Double number line *Spanish: Recta numérica doble*	Two number lines that line up percents with amounts.	0%–100% over 0–300

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

140 tickets; 45% are student tickets. How many?

1. 45% = 0.45
2. 0.45 × 140 = 63

Answer: 63 student tickets

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Find 40% of 80. Step 1 — what is the base? (80, the number after “of”)

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The base is _____ and the percent is _____ %.
- Written in hundredths, _____ % is _____ .
- My equation is part = _____ × _____ .
- The part is _____ , which is smaller than the base because _____ .

Word bank percent | base | whole | part | equation | decimal | hundredths | multiply | double number line | of | check | estimate

WATCH OUT!

A common mistake

A common mistake in Find the Percent of a Number is multiplying by the percent number itself instead of its decimal form

LESSON 4.4

Find and Compare with Percentages

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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1 **Circle the letter of the best answer. Show how you know.**

What is 50% of 84?

(A) 34 (B) 42

(C) 50 (D) 168

STEPS TO TRY
1. 50% is a friendly percent — it means exactly one-half of the number.
2. Finding 50% of a number is the same as dividing that number by 2.
3. Set up 84 ÷ 2, or check with the formula: 0.50 × 84.



0% 50% 100%

WORK

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

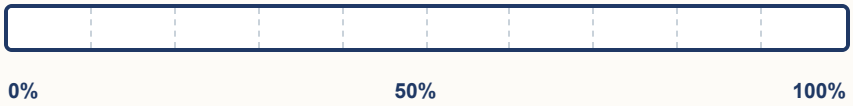
2 **Circle the letter of the best answer. Show how you know.**

What is 10% of 320?

(A) 3.2
(B) 32
(C) 64
(D) 320

STEPS TO TRY

1. 10% means one-tenth — there's a shortcut that needs no multiplication at all.
2. To find 10% of any number, move the decimal point one place to the left (divide by 10).
3. Start with 320 and slide the decimal one place left: 320 → what number?



WORK

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
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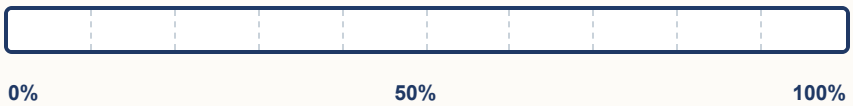
3 **Circle the letter of the best answer. Show how you know.**

What is 20% of 150?

(A) 20
(B) 30
(C) 35
(D) 75

STEPS TO TRY

1. '20% of 150' means you want a PART of 150 — the answer should be smaller than 150.
2. Convert the percent to a decimal, then multiply: Part = Percent × Base.
3. Write 20% as 0.20, then set up 0.20 × 150. (Shortcut: 10% of 150 is 15 — double it.)



WORK

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

4 **Circle the letter of the best answer. Show how you know.**

What is 40% of 50?

(A) 5
(B) 20
(C) 40
(D) 90

STEPS TO TRY

1. 10% is one of 10 equal groups. First find 10% of 50.
2. Divide 50 by 10: 50 ÷ 10 = 5. Now count four groups of 5.
3. 40% has four groups of 10%. Add 5 + 5 + 5 + 5 = 20.



WORK

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

EXPLAIN YOUR THINKING

What steps do you follow every time you find the percent of a number?

SENTENCE STARTER To find ___% of ___, I first convert the percent to ___, then I multiply ___ × ___ = ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help